

Total No. of Questions : 8]

SEAT No. :

P795

[5870]-1109

[Total No. of Pages : 2

T.E. (Chemical)

MASS TRANSFER-I

(2019 Course) (Semester-I) (309341) (Theory)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4 Q.5 or Q.6. Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) 5000 kg/hr of SO₂ air mixture contain 5% by volume SO₂ is to be scrubbed with 2,00,000 kg/hr of water in a packed tower. The exit concentration of SO₂ is reduced to 0.15%. The tower operates at 1 atm. The equilibrium relation is given by $Y = 30 X$

Where $Y = \text{Moles SO}_2 / \text{Moles water}$

$X = \text{Moles SO}_2 / \text{Moles water}$

If the packed height of tower is 420 cm. Estimate the height of transfer unit (H.T.U.) **[6]**

b) Explain Criteri for Selection of Solvent for absorption **[4]**

c) An air-NH₃ mixture with 20 mol % NH₃ is to be scrubbed with water at 30°C and 1 atm. The equipment to be used has an efficiency of 99%. Determine the final concentration of NH₃ in the outgoing gaseous stream. What is the maximum possible solubility of NH₃ in water at this condition? **[8]**

OR

Q2) a) Derive and explain the calculation of number of trays for absorption **[8]**

b) The equilibrium data for a certain absorption may be represented by the equation $Y = 5.1 X^2$ where Y is kmol of gaseous absorbable material per kilomole of inert gas and X is kilomole of dissolved absorbable material per kilomole of inert liquid. The unit is operated as continuous countercurrent absorption. The entering gas contain 5 moles of absorbable material per 20 moles of inert gas. The liquid mixture enters the column as a pure material . Determine the minimum molal ratio of the liquid to gas if the gases leaving the tower contain 1 mole absorbable material per 50 mole of inert gas. **[10]**

P.T.O.

- Q3)** a) What is vapour-liquid equilibria, Explain in Detail with Example. [8]
b) Write a short Note on (Any Three) [9]
i) Cooling towers,
ii) Tray towers,
iii) Spray chambers,
iv) Spray ponds,

OR

- Q4)** a) Discuss about calculation of height of cooling tower. [8]
b) What is Humidification and Dehumidification, Explain in Detail with Example? [6]
c) What is Lewis Relation. [3]
- Q5)** a) Explain in detail about Gas dispersal equipment with neat diagram. [8]
b) What is Trey Efficiency? Explain types of trays and efficiency in Detail. [9]

OR

- Q6)** a) Explain Constriction, Working and Application of mechanically agitated vessels. [8]
b) Discuss about Various Types of Trays and Packing used in Chemical Industry. [9]
- Q7)** a) A dryer reduces the moisture content of 100kg of a potato product from 80% to 10% moisture. 250kg of steam at 70kPa gauge is used to heat 49,800m³ of air to 80°C, and the air is cooled to 71°C in passing through the dryer. Calculate the efficiency of the dryer. The Specific heat of potato is 3.43kJkg⁻¹°C⁻¹. Assume potato enters at 24°C, which is also the ambient air temperature, and leaves at the same temperature as the exit air. [10]
b) Explain in Detail about Role of Mass Transfer in Tray Dryer. Also state the Application of Tray Dryer. [8]

OR

- Q8)** a) Draw and Explain Rate of Drying Curve. [6]
b) A Certain Material was dried under constant drying condition and it was found that 2 hours are required to reduce the free moisture content from 20% to 10%. How much longer would be required to reduce free moisture content to 4%. Assume that no constant rate period is encountered. [6]
c) Write a short note on [6]
i) Pneumatic Dryer and vacuum dryer
ii) Fluidized bed Dryer



Total No. of Questions: 8]

SEAT No. :

PA-1408

[5926]-23

[Total No. of Pages : 3

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(2019 Pattern) (Semester-I) (309341)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain Choice of Solvent for Gas Absorption? **[8]**

- b) A packed tower is designed to recover 98% CO₂ from a gas mixture containing 10% CO₂ & 90% air using water. The equilibrium relation is given as $Y=14X$.

Where $Y=\text{kgCO}_2/\text{kg dry air}$ & $X=\text{kg CO}_2/\text{Kg water}$. The water to gas rate is kept 30% more than the minimum value. Calculate the height of tower if (HTU)_{OG} = 1m. **[9]**

OR

Q2) a) What is Absorption factor? Explain minimum L/G ratio and Stripping factor? **[8]**

- b) 5,000 kg/hr of a SO₂- air mixture containing 5% by volume SO₂ is to be scrubbed with 2,00,000 kg/hr. of water in a packed tower. The exit concentration of SO₂ is reduced to 0.15% The tower operates at 1 atm. The equilibrium relationship is given by $Y=30X$ Where, $Y=\text{Mole SO}_2/\text{Mole air}$ & $X=\text{Mole SO}_2/\text{Mole water}$. If the packed height of the tower is 420 cm, estimate the height of transfer unit (HTU)? **[9]**

Q3) a) Write a short note on Adiabatic Saturation temperature in Humidification operation? **[8]**

P.T.O.

- b) The air in a room is at 26.7°C & a pressure of 101.325KPa & contains water vapor with a partial pressure 2.76 kPa . Calculate [9]
- Absolute Humidity
 - Molal Humidity
 - Saturation humidity
 - Percentage humidity
 - Percentage Relative humidity
 - Relative humidity
 - Humid Heat
 - Humid volume

Data: Vapour pressure of water at 26.7°C is 3.5 KPa

OR

- Q4)** a) Explain HTU, NTU concept in connection with cooling tower? [8]
- b) A gas (B)-Benzene (A) mixture is saturated at 1std.atm 50°C Calculate the absolute humidity if B is [9]
- Nitrogen
 - Carbon dioxide and
 - Oxygen (Data: $P_A=0.362\text{ std. atm.}$)

- Q5)** a) What are the various equipment's used for gas-liquid contact. With neat sketch explain Venturi Scrubber? [9]
- b) Explain the different types of Column in detail with figures? [9]

OR

- Q6)** a) Explain mechanically agitated vessels with different types of impellers [9]
- b) What is Tray efficiency? Explain types of Tray efficiencies? [9]
- Q7)** a) Explain experimental determination of Rate of drying curve? [8]
- b) A porous dry solid was dried under constant drying conditions in a batch dryer. It took 6 hrs to reduce the moisture from 50% to 10% All the moisture content on dry basis. How long will it take to dry a sample of the above solid to dry from 60% to 6% under the same drying conditions? [10]

OR

Q8) a) Explain time required for Drying? Derive the equation to calculate total time for Drying? **[8]**

b) A batch of solid for which the following table of data applies is to be dried from 25% to 6% (wet basis) moisture under conditions identical to those for which the data were tabulated. The initial weight of the wet solid is 200 kg & the drying surface is 1 m²/8 kg dry weight Determine the time for drying? **[10]**

X Kgmoisture/ kg dry solid	0.35	0.25	0.2	0.18	0.16	0.14	0.12	0.10	0.09	0.08	0.064
R Kg moisture hr.m ²	0.3	0.3	0.3	0.266	0.239	0.208	0.18	0.15	0.097	0.07	0.025

